

Stochastic Processes - Homework 4

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Problem 1: Detailed balance of the random-scan Gibbs sampler

Let

$$\Omega = \Omega_1 \times \cdots \times \Omega_d$$

be a finite product space, where each Ω_i is a finite set. Let π be a probability distribution on Ω such that $\pi(x) > 0$ for every $x \in \Omega$.

For a state $x = (x_1, \dots, x_d) \in \Omega$, write

$$x_{-i} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_d)$$

for the collection of all coordinates except the i -th one.

We consider the following random-scan Gibbs sampler: given the current state $X_t = x$,

1. choose an index $I_t \in \{1, \dots, d\}$ uniformly at random;
2. update only the I_t -th coordinate by drawing

$$X_{t+1}^{(I_t)} \sim \pi(\cdot \mid X_t^{(-I_t)}),$$

while keeping all other coordinates unchanged.

In other words, if $I_t = i$, then X_{t+1} is obtained from X_t by resampling the i -th coordinate from the conditional distribution

$$\pi(\cdot \mid x_{-i}).$$

1. Show that if x and y differ in more than one coordinate, then $P(x, y) = 0$.

Solution: Recall that in one step of the random-scan Gibbs sampler, we first choose an index $I_t \in \{1, \dots, d\}$ uniformly at random, and then update only the I_t -th coordinate using the conditional distribution $\pi(\cdot \mid x_{-I_t})$, leaving all other coordinates unchanged.

Let $x, y \in \Omega$ be two states that differ in more than one coordinate. Then for every index $i \in \{1, \dots, d\}$, one of the following holds:

- either $x_j \neq y_j$ for some $j \neq i$, in which case it is impossible to obtain y from x by changing only coordinate i and keeping all other coordinates fixed;
- or $x_i \neq y_i$ and there exists $j \neq i$ with $x_j \neq y_j$, so again y cannot be obtained by changing a single coordinate of x .

Thus, conditional on any choice $I_t = i$, the probability that $X_{t+1} = y$ given $X_t = x$ is zero, because the transition from x to y would require changing more than one coordinate in a single step.

Since the index I_t is the only random choice that determines which coordinate is updated, and for each i the probability of transitioning from x to y in one step is zero, we have

$$P(x, y) = \mathbb{P}(X_{t+1} = y \mid X_t = x) = \sum_{i=1}^d \mathbb{P}(I_t = i) \mathbb{P}(X_{t+1} = y \mid X_t = x, I_t = i) = 0.$$

Hence, if x and y differ in more than one coordinate, then $P(x, y) = 0$, as required.

2. Suppose x and y differ in exactly one coordinate, say the i -th one, and satisfy

$$x_{-i} = y_{-i}.$$

Show that

$$P(x, y) = \frac{1}{d} \pi(y_i \mid x_{-i}).$$

Solution: Suppose $x, y \in \Omega$ differ in exactly one coordinate, say the i -th one, and satisfy $x_{-i} = y_{-i}$. Then the only way to move from x to y in one step of the random-scan Gibbs sampler is:

- first choose $I_t = i$ in step 1, which happens with probability $\mathbb{P}(I_t = i) = 1/d$;
- then, in step 2, resample the i -th coordinate from the conditional distribution $\pi(\cdot \mid x_{-i})$ and obtain the value y_i .

If $I_t \neq i$, the i -th coordinate of X_t is not updated, so we cannot have $X_{t+1} = y$ because $x_i \neq y_i$. Thus,

$$P(x, y) = \mathbb{P}(X_{t+1} = y \mid X_t = x) = \mathbb{P}(I_t = i) \mathbb{P}(X_{t+1} = y \mid X_t = x, I_t = i).$$

Given $X_t = x$ and $I_t = i$, we update only the i -th coordinate using $\pi(\cdot \mid x_{-i})$ and keep all other coordinates fixed at $x_{-i} = y_{-i}$. Therefore,

$$\mathbb{P}(X_{t+1} = y \mid X_t = x, I_t = i) = \mathbb{P}(X_{t+1,i} = y_i \mid X_t = x, I_t = i) = \pi(y_i \mid x_{-i}).$$

Combining these two factors, we obtain

$$P(x, y) = \frac{1}{d} \pi(y_i \mid x_{-i}),$$

as claimed.

3. Prove that the Gibbs sampler satisfies the detailed balance condition with respect to π , namely

$$\pi(x)P(x, y) = \pi(y)P(y, x) \quad \text{for all } x, y \in \Omega.$$

Solution: We need to show that for all $x, y \in \Omega$,

$$\pi(x)P(x, y) = \pi(y)P(y, x).$$

We consider three cases.

Case 1: x and y differ in more than one coordinate. By part (1), $P(x, y) = 0$, and by the same argument with x and y swapped, $P(y, x) = 0$. Hence

$$\pi(x)P(x, y) = \pi(x) \cdot 0 = 0 \quad \text{and} \quad \pi(y)P(y, x) = \pi(y) \cdot 0 = 0,$$

so the detailed balance identity holds.

Case 2: $x = y$. In this case,

$$\pi(x)P(x, x) = \pi(x)P(x, x),$$

so the identity is trivially true.

Case 3: x and y differ in exactly one coordinate. Suppose x and y differ only in the i -th coordinate, and that $x_{-i} = y_{-i}$. By part (2),

$$P(x, y) = \frac{1}{d} \pi(y_i \mid x_{-i}), \quad P(y, x) = \frac{1}{d} \pi(x_i \mid y_{-i}).$$

Since $x_{-i} = y_{-i}$, we can write both conditionals with the same conditioning argument and use the definition of conditional probability:

$$\pi(y_i | x_{-i}) = \frac{\pi(y_i, x_{-i})}{\pi(x_{-i})} = \frac{\pi(y)}{\pi(x_{-i})}, \quad \pi(x_i | y_{-i}) = \frac{\pi(x_i, y_{-i})}{\pi(y_{-i})} = \frac{\pi(x)}{\pi(y_{-i})}.$$

Again using $x_{-i} = y_{-i}$, we obtain

$$\pi(y_i | x_{-i}) = \frac{\pi(y)}{\pi(x_{-i})}, \quad \pi(x_i | y_{-i}) = \frac{\pi(x)}{\pi(x_{-i})}.$$

Therefore,

$$\pi(x)P(x, y) = \pi(x) \frac{1}{d} \pi(y_i | x_{-i}) = \frac{1}{d} \pi(x) \frac{\pi(y)}{\pi(x_{-i})},$$

and

$$\pi(y)P(y, x) = \pi(y) \frac{1}{d} \pi(x_i | y_{-i}) = \frac{1}{d} \pi(y) \frac{\pi(x)}{\pi(x_{-i})}.$$

These two expressions are equal, hence

$$\pi(x)P(x, y) = \pi(y)P(y, x)$$

whenever x and y differ in exactly one coordinate.

Combining the three cases, we conclude that the random-scan Gibbs sampler satisfies the detailed balance condition with respect to π .

4. Conclude that π is a stationary distribution of the random-scan Gibbs sampler.

Solution: From part (3), we know that π satisfies the detailed balance condition with respect to the transition kernel P :

$$\pi(x)P(x, y) = \pi(y)P(y, x) \quad \text{for all } x, y \in \Omega.$$

To show that π is stationary, we must prove that for every $y \in \Omega$,

$$\sum_{x \in \Omega} \pi(x)P(x, y) = \pi(y).$$

Using detailed balance, we have

$$\sum_{x \in \Omega} \pi(x)P(x, y) = \sum_{x \in \Omega} \pi(y)P(y, x) = \pi(y) \sum_{x \in \Omega} P(y, x).$$

Since P is a Markov transition kernel, each row of P sums to one, so

$$\sum_{x \in \Omega} P(y, x) = 1.$$

Therefore,

$$\sum_{x \in \Omega} \pi(x) P(x, y) = \pi(y) \cdot 1 = \pi(y),$$

which shows that π is a stationary distribution of the random-scan Gibbs sampler.

Problem 2: A 2×2 Ising model

Let the state space be

$$\Omega = \{(1, 1), (1, -1), (-1, 1), (-1, -1)\},$$

and define the target distribution

$$\pi_\beta(\sigma) \propto \exp(\beta \sigma_1 \sigma_2), \quad \beta \geq 0,$$

where $\sigma \in \{-1, 1\}^2$.

This is the Ising model on a graph with one edge and no external field. Write

$$\rho \stackrel{\text{def}}{=} \tanh \beta.$$

1. Calculate $P(\sigma_1 \mid \sigma_2 = 1)$ and $P(\sigma_1 \mid \sigma_2 = -1)$.

Solution: The target distribution on $\sigma = (\sigma_1, \sigma_2) \in \{-1, 1\}^2$ is

$$\pi_\beta(\sigma) \propto \exp(\beta \sigma_1 \sigma_2).$$

To compute the conditional distribution of σ_1 given σ_2 , we treat σ_2 as fixed and compare the two possible values $\sigma_1 \in \{-1, 1\}$.

Case $\sigma_2 = 1$. Then for $\sigma_1 = 1$,

$$\pi_\beta(\sigma_1 = 1 \mid \sigma_2 = 1) \propto \exp(\beta \cdot 1 \cdot 1) = e^\beta,$$

and for $\sigma_1 = -1$,

$$\pi_\beta(\sigma_1 = -1 \mid \sigma_2 = 1) \propto \exp(\beta \cdot (-1) \cdot 1) = e^{-\beta}.$$

Normalizing,

$$\mathbb{P}(\sigma_1 = 1 \mid \sigma_2 = 1) = \frac{e^\beta}{e^\beta + e^{-\beta}} = \frac{e^{2\beta}}{e^{2\beta} + 1}, \quad \mathbb{P}(\sigma_1 = -1 \mid \sigma_2 = 1) = \frac{e^{-\beta}}{e^\beta + e^{-\beta}} = \frac{1}{e^{2\beta} + 1}.$$

Case $\sigma_2 = -1$. Then for $\sigma_1 = 1$,

$$\pi_\beta(\sigma_1 = 1 \mid \sigma_2 = -1) \propto \exp(\beta \cdot 1 \cdot (-1)) = e^{-\beta},$$

and for $\sigma_1 = -1$,

$$\pi_\beta(\sigma_1 = -1 \mid \sigma_2 = -1) \propto \exp(\beta \cdot (-1) \cdot (-1)) = e^\beta.$$

Normalizing,

$$\mathbb{P}(\sigma_1 = 1 \mid \sigma_2 = -1) = \frac{e^{-\beta}}{e^\beta + e^{-\beta}} = \frac{1}{e^{2\beta} + 1}, \quad \mathbb{P}(\sigma_1 = -1 \mid \sigma_2 = -1) = \frac{e^\beta}{e^\beta + e^{-\beta}} = \frac{e^{2\beta}}{e^{2\beta} + 1}.$$

Equivalently, in terms of $\rho = \tanh \beta$, these can be written as

$$\mathbb{P}(\sigma_1 = s \mid \sigma_2 = t) = \frac{1}{2}(1 + st\rho), \quad s, t \in \{-1, 1\}.$$

2. Write down the 4×4 transition matrix for the (random-scan) Gibbs sampler for this model.

Solution: We use the random-scan Gibbs sampler which, at each step, chooses one of the two coordinates uniformly at random and updates it from its conditional distribution given the other coordinate.

Let us fix the state ordering

$$s_1 = (1, 1), \quad s_2 = (1, -1), \quad s_3 = (-1, 1), \quad s_4 = (-1, -1).$$

We also write the conditional probabilities from part (a) as

$$\mathbb{P}(\sigma_1 = s \mid \sigma_2 = t) = \frac{1}{2}(1 + st\rho), \quad \mathbb{P}(\sigma_2 = t \mid \sigma_1 = s) = \frac{1}{2}(1 + st\rho),$$

for $s, t \in \{-1, 1\}$, where $\rho = \tanh \beta$.

For each current state x , the next state y is obtained by:

- with probability $1/2$, updating σ_1 given σ_2 ;
- with probability $1/2$, updating σ_2 given σ_1 .

We now compute each row of the 4×4 transition matrix P .

From $s_1 = (1, 1)$.

- Update σ_1 (prob. $1/2$): given $\sigma_2 = 1$,

$$\mathbb{P}(\sigma_1 = 1 \mid \sigma_2 = 1) = \frac{1 + \rho}{2}, \quad \mathbb{P}(\sigma_1 = -1 \mid \sigma_2 = 1) = \frac{1 - \rho}{2}.$$

Thus,

$$\mathbb{P}((1,1) \rightarrow (1,1) \text{ via } \sigma_1) = \frac{1+\rho}{2}, \quad \mathbb{P}((1,1) \rightarrow (-1,1) \text{ via } \sigma_1) = \frac{1-\rho}{2}.$$

- Update σ_2 (prob. 1/2): given $\sigma_1 = 1$,

$$\mathbb{P}(\sigma_2 = 1 \mid \sigma_1 = 1) = \frac{1+\rho}{2}, \quad \mathbb{P}(\sigma_2 = -1 \mid \sigma_1 = 1) = \frac{1-\rho}{2}.$$

Thus,

$$\mathbb{P}((1,1) \rightarrow (1,1) \text{ via } \sigma_2) = \frac{1+\rho}{2}, \quad \mathbb{P}((1,1) \rightarrow (1,-1) \text{ via } \sigma_2) = \frac{1-\rho}{2}.$$

Combining the two possibilities (each with weight 1/2),

$$P_{11} = \frac{1}{2} \cdot \frac{1+\rho}{2} + \frac{1}{2} \cdot \frac{1+\rho}{2} = \frac{1+\rho}{2},$$

$$P_{12} = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1-\rho}{2} = \frac{1-\rho}{4}, \quad P_{13} = \frac{1}{2} \cdot \frac{1-\rho}{2} + \frac{1}{2} \cdot 0 = \frac{1-\rho}{4}, \quad P_{14} = 0.$$

From $s_2 = (1, -1)$.

- Update σ_1 (prob. 1/2): given $\sigma_2 = -1$,

$$\mathbb{P}(\sigma_1 = 1 \mid \sigma_2 = -1) = \frac{1-\rho}{2}, \quad \mathbb{P}(\sigma_1 = -1 \mid \sigma_2 = -1) = \frac{1+\rho}{2}.$$

Thus,

$$(1, -1) \rightarrow (1, -1) \text{ via } \sigma_1 \text{ with prob } \frac{1-\rho}{2}, \quad (1, -1) \rightarrow (-1, -1) \text{ with prob } \frac{1+\rho}{2}.$$

- Update σ_2 (prob. 1/2): given $\sigma_1 = 1$,

$$\mathbb{P}(\sigma_2 = -1 \mid \sigma_1 = 1) = \frac{1-\rho}{2}, \quad \mathbb{P}(\sigma_2 = 1 \mid \sigma_1 = 1) = \frac{1+\rho}{2}.$$

Thus,

$$(1, -1) \rightarrow (1, -1) \text{ via } \sigma_2 \text{ with prob } \frac{1-\rho}{2}, \quad (1, -1) \rightarrow (1, 1) \text{ with prob } \frac{1+\rho}{2}.$$

Therefore,

$$P_{22} = \frac{1}{2} \cdot \frac{1-\rho}{2} + \frac{1}{2} \cdot \frac{1-\rho}{2} = \frac{1-\rho}{2},$$

$$P_{21} = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1+\rho}{2} = \frac{1+\rho}{4}, \quad P_{24} = \frac{1}{2} \cdot \frac{1+\rho}{2} + \frac{1}{2} \cdot 0 = \frac{1+\rho}{4}, \quad P_{23} = 0.$$

From $s_3 = (-1, 1)$. By symmetry with s_2 , we obtain

$$P_{33} = \frac{1-\rho}{2}, \quad P_{31} = \frac{1+\rho}{4}, \quad P_{34} = \frac{1+\rho}{4}, \quad P_{32} = 0.$$

From $s_4 = (-1, -1)$. By symmetry with s_1 , we obtain

$$P_{44} = \frac{1+\rho}{2}, \quad P_{42} = \frac{1-\rho}{4}, \quad P_{43} = \frac{1-\rho}{4}, \quad P_{41} = 0.$$

Collecting all rows, the 4×4 transition matrix in the state order $\{(1, 1), (1, -1), (-1, 1), (-1, -1)\}$ is

$$P = \begin{pmatrix} \frac{1+\rho}{2} & \frac{1-\rho}{4} & \frac{1-\rho}{4} & 0 \\ \frac{1+\rho}{4} & \frac{1-\rho}{2} & 0 & \frac{1+\rho}{4} \\ \frac{1+\rho}{4} & 0 & \frac{1-\rho}{2} & \frac{1+\rho}{4} \\ 0 & \frac{1-\rho}{4} & \frac{1-\rho}{4} & \frac{1+\rho}{2} \end{pmatrix}.$$

3. Calculate the eigenvalues and corresponding eigenvectors of this transition matrix, then calculate the spectral gap. Show that the spectral gap goes to 0 as $\beta \rightarrow \infty$.

Solution: We use the state order

$$s_1 = (1, 1), \quad s_2 = (1, -1), \quad s_3 = (-1, 1), \quad s_4 = (-1, -1)$$

and the transition matrix from part (b),

$$P = \begin{pmatrix} \frac{1+\rho}{2} & \frac{1-\rho}{4} & \frac{1-\rho}{4} & 0 \\ \frac{1+\rho}{4} & \frac{1-\rho}{2} & 0 & \frac{1+\rho}{4} \\ \frac{1+\rho}{4} & 0 & \frac{1-\rho}{2} & \frac{1+\rho}{4} \\ 0 & \frac{1-\rho}{4} & \frac{1-\rho}{4} & \frac{1+\rho}{2} \end{pmatrix}.$$

By symmetry, it is convenient to work in the orthogonal basis of functions on Ω :

$$f_0(\sigma) = 1, \quad f_1(\sigma) = \sigma_1 + \sigma_2, \quad f_2(\sigma) = \sigma_1 - \sigma_2, \quad f_3(\sigma) = \sigma_1 \sigma_2.$$

In the chosen state order these correspond to vectors

$$v_0 = (1, 1, 1, 1)^\top, \quad v_1 = (2, 0, 0, -2)^\top, \quad v_2 = (0, 2, -2, 0)^\top, \quad v_3 = (1, -1, -1, 1)^\top.$$

Eigenvalue 1. Since P is a Markov transition matrix, the constant function is an eigenfunction with eigenvalue 1:

$$Pv_0 = v_0,$$

so $\lambda_0 = 1$.

Eigenvalues for f_1 and f_2 . Let $f_1(\sigma) = \sigma_1 + \sigma_2$. Under one random-scan Gibbs step, we update one spin (chosen uniformly) from its conditional distribution. Using

$$\mathbb{E}[\sigma'_1 \mid \sigma_2] = \rho \sigma_2, \quad \mathbb{E}[\sigma'_2 \mid \sigma_1] = \rho \sigma_1,$$

and the fact that we update each spin with probability $1/2$, we obtain

$$\mathbb{E}[\sigma'_1 \mid \sigma] = \frac{1}{2}\sigma_1 + \frac{1}{2}\rho\sigma_2, \quad \mathbb{E}[\sigma'_2 \mid \sigma] = \frac{1}{2}\rho\sigma_1 + \frac{1}{2}\sigma_2.$$

Hence

$$\mathbb{E}[f_1(\sigma') \mid \sigma] = \mathbb{E}[\sigma'_1 + \sigma'_2 \mid \sigma] = \frac{1+\rho}{2}(\sigma_1 + \sigma_2) = \frac{1+\rho}{2}f_1(\sigma),$$

so f_1 is an eigenfunction with eigenvalue

$$\lambda_1 = \frac{1+\rho}{2}.$$

By symmetry of the model, the same calculation holds for $f_2(\sigma) = \sigma_1 - \sigma_2$, and we obtain another eigenvalue

$$\lambda_2 = \frac{1+\rho}{2}.$$

Eigenvalue for f_3 . For $f_3(\sigma) = \sigma_1\sigma_2$, a direct matrix multiplication (or a short calculation of $\mathbb{E}[\sigma'_1\sigma'_2 \mid \sigma]$) shows that

$$Pv_3 = 0 \cdot v_3,$$

so

$$\lambda_3 = 0.$$

Spectral gap and limit $\beta \rightarrow \infty$. Thus the four eigenvalues of P are

$$\lambda_0 = 1, \quad \lambda_1 = \lambda_2 = \frac{1+\rho}{2}, \quad \lambda_3 = 0.$$

The spectral gap is defined as

$$\gamma = 1 - \max\{|\lambda| : \lambda \neq 1\} = 1 - \max\left\{\frac{1+\rho}{2}, 0\right\} = 1 - \frac{1+\rho}{2} = \frac{1-\rho}{2}.$$

Since $\rho = \tanh \beta \rightarrow 1$ as $\beta \rightarrow \infty$, we have

$$\gamma(\beta) = \frac{1 - \tanh \beta}{2} \rightarrow 0 \quad \text{as } \beta \rightarrow \infty.$$

Thus the spectral gap goes to 0 as $\beta \rightarrow \infty$.

4. Implement the Gibbs sampler yourself (without using LLMs). Run it for 100 iterations for several choices of β , and plot: (1) the sample path, and (2) the histogram of the first coordinate.

Solution: I implemented the random-scan Gibbs sampler described in parts (a)–(b) in Python. At each iteration $t = 0, \dots, 99$ I:

(a) Chose a coordinate $k_t \in \{1, 2\}$ uniformly at random.

(b) If $k_t = 1$, updated σ_1 given $\sigma_2^{(t)}$ using

$$\mathbb{P}(\sigma_1^{(t+1)} = s \mid \sigma_2^{(t)} = t) = \frac{1}{2}(1 + st\rho), \quad s, t \in \{-1, 1\},$$

and set $\sigma_2^{(t+1)} = \sigma_2^{(t)}$.

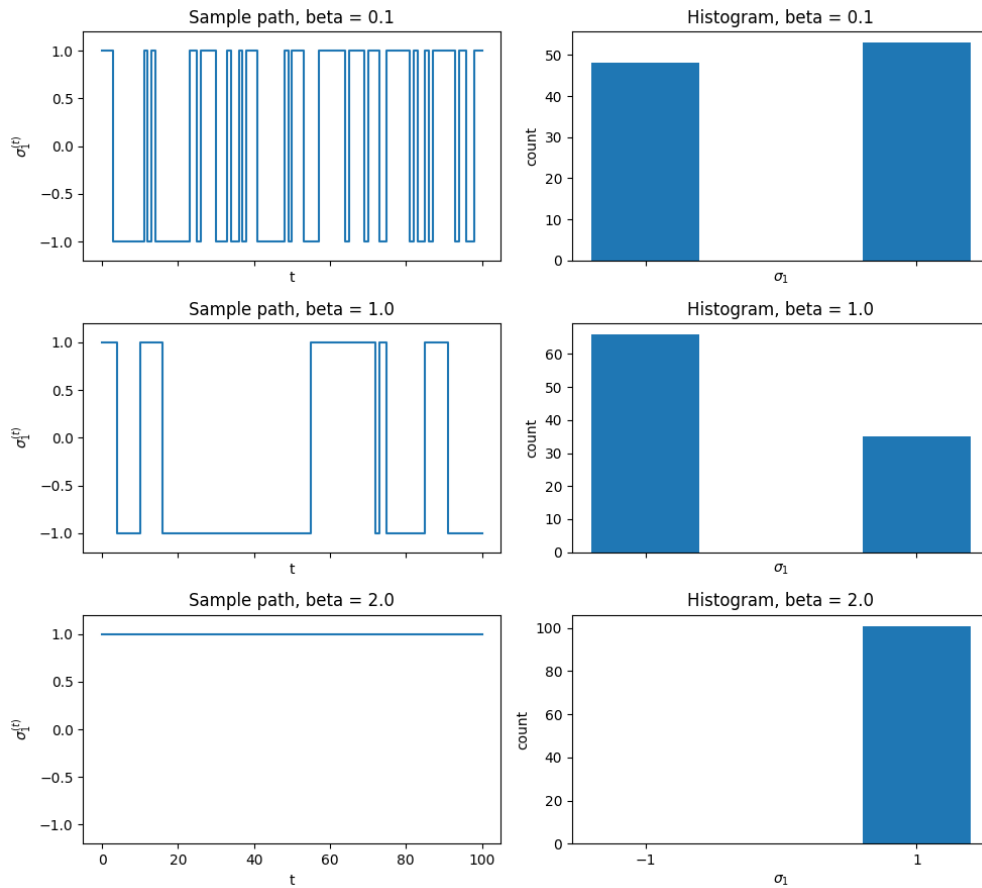
(c) If $k_t = 2$, updated σ_2 given $\sigma_1^{(t)}$ in the same way and kept σ_1 fixed.

The core of the implementation is:

```
def gibbs_step(state, beta, rng):
    sigma1, sigma2 = state
    rho = np.tanh(beta)
    if rng.random() < 0.5:
        p_plus = 0.5 * (1 + rho * sigma2)
        sigma1 = 1 if rng.random() < p_plus else -1
    else:
        p_plus = 0.5 * (1 + rho * sigma1)
        sigma2 = 1 if rng.random() < p_plus else -1
    return (sigma1, sigma2)
```

Starting from $(\sigma_1^{(0)}, \sigma_2^{(0)}) = (1, 1)$, I ran the chain for $T = 100$ iterations for three values of β : 0.1, 1.0, and 2.0. For each run I recorded the first coordinate $(\sigma_1^{(t)})_{t=0}^{100}$ and produced a sample path plot and a histogram:

- For $\beta = 0.1$, the sample path of $\sigma_1^{(t)}$ flips very frequently between $+1$ and -1 , and the histogram is close to balanced between the two values.
- For $\beta = 1.0$, the path shows noticeably longer stretches at each value (blocks of consecutive $+1$ or -1), and the histogram puts more mass on -1 than on $+1$ in this particular run.
- For $\beta = 2.0$, the chain remains at $+1$ for all 100 iterations from the chosen starting point, so the sample path is flat at $+1$ and the histogram has all its mass at $+1$.



These empirical results are consistent with the theoretical spectral gap from part (c): as β increases (so $\rho = \tanh \beta \rightarrow 1$), the chain mixes more slowly, resulting in longer runs and, for large β , very slow movement away from the initial high-probability state.

Problem 3: Gibbs sampler on a continuous state space

Let $x_0, x_1, \dots, x_n \in \mathbb{R}^d$. Suppose

$$x_0 \sim p(x_0),$$

where p is an arbitrary probability density on $\mathbb{R}^d$. For $i = 0, 1, \dots, n-1$, suppose the forward diffusion process is given by

$$x_{i+1} = a_i x_i + \xi_i, \quad \xi_i \sim N(0, \sigma_i^2 I_d),$$

independently across i . We observe

$$y = x_0 + \varepsilon, \quad \varepsilon \sim N(0, \tau^2 I_d),$$

independently of everything else.

1. Show that the joint density of $(x_0, \dots, x_n, y)$ is, up to a constant,

$$p(x_0) \exp\left(-\frac{\|y - x_0\|^2}{2\tau^2}\right) \prod_{i=0}^{n-1} \exp\left(-\frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right).$$

Solution: We are given the hierarchical model

$$x_0 \sim p(x_0), \quad x_{i+1} \mid x_i \sim N(a_i x_i, \sigma_i^2 I_d), \quad i = 0, \dots, n-1,$$

independently across i , and

$$y \mid x_0 \sim N(x_0, \tau^2 I_d),$$

independently of everything else.

By the chain rule for densities, the joint density of $(x_0, \dots, x_n, y)$ can be written as

$$p(x_0, \dots, x_n, y) = p(x_0) \prod_{i=0}^{n-1} p(x_{i+1} \mid x_i) p(y \mid x_0).$$

Each conditional distribution is Gaussian with density

$$p(x_{i+1} \mid x_i) = \frac{1}{(2\pi\sigma_i^2)^{d/2}} \exp\left(-\frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right),$$

and

$$p(y \mid x_0) = \frac{1}{(2\pi\tau^2)^{d/2}} \exp\left(-\frac{\|y - x_0\|^2}{2\tau^2}\right).$$

Therefore,

$$p(x_0, \dots, x_n, y) = p(x_0) \left[\prod_{i=0}^{n-1} \frac{1}{(2\pi\sigma_i^2)^{d/2}} \right] \frac{1}{(2\pi\tau^2)^{d/2}} \exp\left(-\frac{\|y - x_0\|^2}{2\tau^2} - \sum_{i=0}^{n-1} \frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right).$$

All multiplicative constants in front of the exponential can be absorbed into a single proportionality constant. Hence, up to a constant factor,

$$p(x_0, \dots, x_n, y) \propto p(x_0) \exp\left(-\frac{\|y - x_0\|^2}{2\tau^2}\right) \prod_{i=0}^{n-1} \exp\left(-\frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right),$$

which is the desired expression.

2. Assuming we have observed y , we are interested in sampling from the posterior density

$$\pi(x_0, \dots, x_n \mid y)$$

using a Gibbs sampler.

- (a) Fix $i \in \{1, \dots, n-1\}$. Show that the full conditional density of x_i given all the other variables and y ,

$$\pi(x_i \mid x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n, y),$$

is Gaussian. (Hint: The conditional distribution of x_i only depends on x_{i-1} and x_{i+1} . Then try to complete the square.)

Solution: Fix $i \in \{1, \dots, n-1\}$. We consider the full conditional density of x_i given all other variables and y :

$$\pi(x_i \mid x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n, y).$$

From the joint density in the previous part, we know that, up to a constant,

$$p(x_0, \dots, x_n, y) \propto p(x_0) \exp\left(-\frac{\|y - x_0\|^2}{2\tau^2}\right) \prod_{k=0}^{n-1} \exp\left(-\frac{\|x_{k+1} - a_k x_k\|^2}{2\sigma_k^2}\right).$$

In this product, the only factors that depend on x_i are those with $k = i-1$ and $k = i$, namely

$$\exp\left(-\frac{\|x_i - a_{i-1}x_{i-1}\|^2}{2\sigma_{i-1}^2}\right) \quad \text{and} \quad \exp\left(-\frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right).$$

All other terms in the joint density are constant with respect to x_i and can be absorbed into the proportionality constant. Therefore the full conditional density of x_i is, up to a normalizing constant,

$$\pi(x_i \mid \text{rest}) \propto \exp\left(-\frac{\|x_i - a_{i-1}x_{i-1}\|^2}{2\sigma_{i-1}^2} - \frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right).$$

We now show this is a multivariate normal density in x_i by completing the square. Expand the two quadratic terms:

$$\|x_i - a_{i-1}x_{i-1}\|^2 = x_i^\top x_i - 2x_i^\top a_{i-1}x_{i-1} + \|a_{i-1}x_{i-1}\|^2,$$

$$\|x_{i+1} - a_i x_i\|^2 = x_{i+1}^\top x_{i+1} - 2x_{i+1}^\top a_i x_i + x_i^\top a_i^2 x_i.$$

Discarding terms that do not depend on x_i , we obtain that the exponent is

$$-\frac{1}{2} \left[x_i^\top \left(\frac{1}{\sigma_{i-1}^2} I_d + \frac{a_i^2}{\sigma_i^2} I_d \right) x_i - 2x_i^\top \left(\frac{a_{i-1}}{\sigma_{i-1}^2} x_{i-1} + \frac{a_i}{\sigma_i^2} x_{i+1} \right) \right] + \text{const},$$

where “const” denotes terms not depending on x_i . Define

$$\Lambda_i = \left(\frac{1}{\sigma_{i-1}^2} + \frac{a_i^2}{\sigma_i^2} \right) I_d, \quad b_i = \frac{a_{i-1}}{\sigma_{i-1}^2} x_{i-1} + \frac{a_i}{\sigma_i^2} x_{i+1}.$$

Then the exponent can be written as

$$-\frac{1}{2} (x_i^\top \Lambda_i x_i - 2x_i^\top b_i) + \text{const} = -\frac{1}{2} (x_i - m_i)^\top \Lambda_i (x_i - m_i) + \text{const}',$$

where

$$m_i = \Lambda_i^{-1} b_i = \frac{\frac{a_{i-1}}{\sigma_{i-1}^2} x_{i-1} + \frac{a_i}{\sigma_i^2} x_{i+1}}{\frac{1}{\sigma_{i-1}^2} + \frac{a_i^2}{\sigma_i^2}},$$

and the covariance matrix is

$$\Sigma_i = \Lambda_i^{-1} = \left(\frac{1}{\sigma_{i-1}^2} + \frac{a_i^2}{\sigma_i^2} \right)^{-1} I_d.$$

Thus, the full conditional distribution of x_i given all other variables and y is multivariate Gaussian:

$$x_i \mid x_{i-1}, x_{i+1}, \text{rest} \sim N(m_i, \Sigma_i),$$

with mean m_i and covariance Σ_i as above. In particular, the conditional depends on the other variables only through x_{i-1} and x_{i+1} , as claimed.

(b) Show similarly that the last coordinate also has a Gaussian full conditional:

$$x_n \mid x_0, \dots, x_{n-1}, y \sim N(a_{n-1}x_{n-1}, \sigma_{n-1}^2 I_d).$$

Solution: We consider the full conditional distribution of the last state x_n given all other variables and y :

$$\pi(x_n \mid x_0, \dots, x_{n-1}, y).$$

From the joint density (up to a constant),

$$p(x_0, \dots, x_n, y) \propto p(x_0) \exp\left(-\frac{\|y - x_0\|^2}{2\tau^2}\right) \prod_{i=0}^{n-1} \exp\left(-\frac{\|x_{i+1} - a_i x_i\|^2}{2\sigma_i^2}\right),$$

we see that x_n appears only in the last transition factor with index $i = n - 1$:

$$\exp\left(-\frac{\|x_n - a_{n-1}x_{n-1}\|^2}{2\sigma_{n-1}^2}\right).$$

All other factors in the joint density do not depend on x_n and are therefore constants with respect to x_n . Hence the full conditional is, up to a normalizing constant,

$$\pi(x_n \mid x_0, \dots, x_{n-1}, y) \propto \exp\left(-\frac{\|x_n - a_{n-1}x_{n-1}\|^2}{2\sigma_{n-1}^2}\right).$$

This is exactly the kernel of a multivariate normal distribution with mean $a_{n-1}x_{n-1}$ and covariance $\sigma_{n-1}^2 I_d$. Therefore,

$$x_n \mid x_0, \dots, x_{n-1}, y \sim N(a_{n-1}x_{n-1}, \sigma_{n-1}^2 I_d),$$

as claimed.

Problem 4: Independent Metropolis–Hastings

Let $\Omega = \{1, 2, \dots, n\}$ be a finite state space. Suppose the target distribution is

$$\pi = (\pi_1, \dots, \pi_n),$$

and suppose we are able to sample from another distribution

$$p = (p_1, \dots, p_n),$$

where $\pi_i > 0$ and $p_i > 0$ for every i .

Consider the following independent Metropolis–Hastings sampler. If the current state is $X_t = i$, then:

1. propose a new state J from the distribution p , independently of the current state;
2. accept the proposal with probability

$$\alpha(i, J) = \min \left(1, \frac{\pi_J p_i}{\pi_i p_J} \right);$$

otherwise, stay at i .

Define

$$w_i = \frac{\pi_i}{p_i}, \quad i = 1, \dots, n,$$

and assume, without loss of generality, that

$$w_1 \geq w_2 \geq \dots \geq w_n.$$

1. Show that for $i \neq j$,

$$P(i, j) = \begin{cases} p_j, & \text{if } j < i, \\ \frac{p_j}{w_j} w_i, & \text{if } j > i. \end{cases}$$

Also write down a formula for $P(i, i)$.

Solution: Recall the independent Metropolis–Hastings update: if $X_t = i$, we propose $J \sim p$ independently of i and accept the proposal with probability

$$\alpha(i, J) = \min \left(1, \frac{\pi_J p_i}{\pi_i p_J} \right) = \min \left(1, \frac{w_J}{w_i} \right),$$

where $w_k = \pi_k/p_k$ for $k = 1, \dots, n$. Thus the one-step transition probabilities are

$$P(i, j) = \mathbb{P}(X_{t+1} = j \mid X_t = i) = \begin{cases} p_j \alpha(i, j), & j \neq i, \\ 1 - \sum_{k \neq i} p_k \alpha(i, k), & j = i. \end{cases}$$

We now use the ordering assumption $w_1 \geq w_2 \geq \dots \geq w_n$ to simplify $\alpha(i, j)$ according to whether $j < i$ or $j > i$.

Case $j < i$. Since the weights are nonincreasing, $j < i$ implies $w_j \geq w_i$. Hence

$$\frac{w_j}{w_i} \geq 1 \implies \alpha(i, j) = \min \left(1, \frac{w_j}{w_i} \right) = 1.$$

Therefore, for $j < i$,

$$P(i, j) = p_j \alpha(i, j) = p_j.$$

Case $j > i$. Now $j > i$ implies $w_j \leq w_i$, so

$$\frac{w_j}{w_i} \leq 1 \implies \alpha(i, j) = \min\left(1, \frac{w_j}{w_i}\right) = \frac{w_j}{w_i}.$$

Thus for $j > i$,

$$P(i, j) = p_j \alpha(i, j) = p_j \frac{w_j}{w_i}.$$

Combining the two cases, for $i \neq j$ we have

$$P(i, j) = \begin{cases} p_j, & j < i, \\ p_j \frac{w_j}{w_i}, & j > i, \end{cases}$$

as required.

Formula for $P(i, i)$. Finally, the probability of staying at i is one minus the probability of moving to any other state:

$$P(i, i) = 1 - \sum_{j \neq i} P(i, j) = 1 - \sum_{j < i} p_j - \sum_{j > i} p_j \frac{w_j}{w_i}.$$

This gives an explicit expression for $P(i, i)$ in terms of $\{p_j, w_j\}$.

2. Show that

$$P = U + R,$$

where R is a rank-one matrix and U is upper triangular. In particular, show that the diagonal entries of U are

$$U_{ii} = \sum_{j > i} p_j \left(1 - \frac{w_j}{w_i}\right), \quad i = 1, \dots, n.$$

(Hint: Try subtracting the vector $(p_1, p_2, \dots, p_n)$ from every row of P .)

Solution: From part (1), for $i \neq j$ we have

$$P(i, j) = \begin{cases} p_j, & j < i, \\ p_j \frac{w_j}{w_i}, & j > i, \end{cases}$$

and

$$P(i, i) = 1 - \sum_{j < i} p_j - \sum_{j > i} p_j \frac{w_j}{w_i}.$$

Decomposition $P = U + R$. Define the matrix R to be the rank-one matrix

$$R_{ij} = p_j, \quad i, j = 1, \dots, n.$$

That is, R has the same row vector $(p_1, \dots, p_n)$ in every row. Then define

$$U = P - R.$$

By construction, R clearly has rank one (it is the outer product of the all-ones vector with the row vector $(p_1, \dots, p_n)$), and we have $P = U + R$.

We now compute the entries of U . For $i \neq j$,

$$U(i, j) = P(i, j) - p_j = \begin{cases} p_j - p_j = 0, & j < i, \\ p_j \frac{w_j}{w_i} - p_j = p_j \left(\frac{w_j}{w_i} - 1 \right) = -p_j \left(1 - \frac{w_j}{w_i} \right), & j > i. \end{cases}$$

Thus $U(i, j) = 0$ whenever $j < i$, so U is upper triangular. For $j > i$, the entries $U(i, j)$ may be nonzero, but they appear only above the diagonal.

For the diagonal entries,

$$U(i, i) = P(i, i) - p_i = \left(1 - \sum_{j < i} p_j - \sum_{j > i} p_j \frac{w_j}{w_i} \right) - p_i.$$

Rewriting $1 - \sum_{j < i} p_j - p_i$ as $\sum_{j > i} p_j$, we obtain

$$U(i, i) = \sum_{j > i} p_j - \sum_{j > i} p_j \frac{w_j}{w_i} = \sum_{j > i} p_j \left(1 - \frac{w_j}{w_i} \right).$$

We have therefore decomposed P as

$$P = U + R,$$

where R is rank one, U is upper triangular (since $U(i, j) = 0$ for $j < i$), and the diagonal entries of U are

$$U_{ii} = \sum_{j > i} p_j \left(1 - \frac{w_j}{w_i} \right), \quad i = 1, \dots, n,$$

as required.

3. (Rank-one perturbation) Suppose A is an $n \times n$ real matrix with real eigenvalues

$$\lambda_1, \dots, \lambda_n$$

and corresponding real eigenvectors

$$u_1, \dots, u_n.$$

Let $u \in \mathbb{R}^n$ be any vector such that

$$\lambda_i - \lambda_n - u^\top u_n \neq 0 \quad \text{for all } i = 1, \dots, n-1.$$

Define

$$B = A + u_n u^\top.$$

Then B has eigenvalues

$$\lambda_1, \lambda_2, \dots, \lambda_{n-1}, \lambda_n + u^\top u_n,$$

with corresponding eigenvectors

$$\tilde{u}_j = \begin{cases} u_n, & \text{if } j = n, \\ u_j + c_j u_n, & \text{if } j \neq n, \end{cases}$$

where

$$c_j = \frac{u^\top u_j}{\lambda_j - \lambda_n - u^\top u_n}.$$

Solution: Let A be an $n \times n$ real matrix with real eigenvalues $\lambda_1, \dots, \lambda_n$ and corresponding eigenvectors $u_1, \dots, u_n$. Let $u \in \mathbb{R}^n$ be such that

$$\lambda_i - \lambda_n - u^\top u_n \neq 0 \quad \text{for all } i = 1, \dots, n-1.$$

Define the rank-one perturbation

$$B = A + u_n u^\top.$$

We show that B has eigenvalues $\lambda_1, \dots, \lambda_{n-1}, \lambda_n + u^\top u_n$ with corresponding eigenvectors

$$\tilde{u}_j = \begin{cases} u_n, & j = n, \\ u_j + c_j u_n, & j \neq n, \end{cases} \quad c_j = \frac{u^\top u_j}{\lambda_j - \lambda_n - u^\top u_n}.$$

Case $j = n$. Take $\tilde{u}_n = u_n$. Then

$$B\tilde{u}_n = (A + u_n u^\top)u_n = Au_n + u_n(u^\top u_n) = \lambda_n u_n + (u^\top u_n)u_n = (\lambda_n + u^\top u_n)u_n.$$

Thus u_n is an eigenvector of B with eigenvalue $\lambda_n + u^\top u_n$.

Case $j \neq n$. For $j \neq n$, define

$$\tilde{u}_j = u_j + c_j u_n,$$

with c_j to be chosen so that $\tilde{u}_j$ is an eigenvector of B with eigenvalue λ_j .

Compute $B\tilde{u}_j$:

$$\begin{aligned} B\tilde{u}_j &= (A + u_n u^\top)(u_j + c_j u_n) \\ &= Au_j + c_j Au_n + u_n u^\top u_j + c_j u_n u^\top u_n \\ &= \lambda_j u_j + c_j \lambda_n u_n + u_n(u^\top u_j) + c_j u_n(u^\top u_n). \end{aligned}$$

Group terms in the basis $\{u_j, u_n\}$:

$$B\tilde{u}_j = \lambda_j u_j + u_n(c_j \lambda_n + u^\top u_j + c_j u^\top u_n).$$

On the other hand, if $\tilde{u}_j$ is an eigenvector of B with eigenvalue λ_j , then

$$B\tilde{u}_j = \lambda_j \tilde{u}_j = \lambda_j(u_j + c_j u_n) = \lambda_j u_j + c_j \lambda_j u_n.$$

Comparing coefficients of u_j and u_n gives:

$$c_j \lambda_n + u^\top u_j + c_j u^\top u_n = c_j \lambda_j.$$

Rearranging,

$$u^\top u_j = c_j(\lambda_j - \lambda_n - u^\top u_n),$$

and hence

$$c_j = \frac{u^\top u_j}{\lambda_j - \lambda_n - u^\top u_n}.$$

By assumption, the denominator is nonzero for all $j = 1, \dots, n-1$, so each c_j is well-defined. Therefore, for every $j \neq n$, the vector $\tilde{u}_j = u_j + c_j u_n$ is an eigenvector of B with eigenvalue λ_j .

Combining the two cases, B has eigenvalues $\lambda_1, \dots, \lambda_{n-1}, \lambda_n + u^\top u_n$ with corresponding eigenvectors as claimed.

4. Apply this fact to the decomposition

$$P = U + R$$

from the previous part, where R is rank one, to derive the eigenvalues of P . (Hint: The eigenvalues of an upper triangular matrix are precisely its diagonal entries.)

Solution: From part (2), we have decomposed the transition matrix as

$$P = U + R,$$

where R is rank one and U is upper triangular with diagonal entries

$$U_{ii} = \sum_{j>i} p_j \left(1 - \frac{w_j}{w_i}\right), \quad i = 1, \dots, n.$$

Since U is upper triangular, its eigenvalues are precisely its diagonal entries:

$$\lambda_i(U) = U_{ii}, \quad i = 1, \dots, n.$$

Next we identify R in the form required by the rank-one perturbation lemma. By definition,

$$R_{ij} = p_j, \quad i, j = 1, \dots, n,$$

so every row of R is equal to the row vector $(p_1, \dots, p_n)$. Equivalently,

$$R = \mathbf{1} p^\top,$$

where $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^n$ and $p = (p_1, \dots, p_n)^\top$. Thus R has rank one and can be written in the form

$$R = u_n u^\top \quad \text{with} \quad u_n = \mathbf{1}, \quad u = p.$$

We now apply the rank-one perturbation lemma from part (3) with

$$A = U, \quad B = P = U + R, \quad u_n = \mathbf{1}, \quad u = p.$$

The lemma states that P has eigenvalues

$$\lambda_1(U), \dots, \lambda_{n-1}(U), \quad \lambda_n(U) + u^\top u_n,$$

i.e. all but one of the eigenvalues of U remain unchanged, and a single eigenvalue is shifted by the amount $u^\top u_n$.

Because P is a Markov transition matrix, it must have eigenvalue 1 with (right) eigenvector equal to the stationary distribution (up to scaling). Hence the shifted eigenvalue

$$\lambda_n(U) + u^\top u_n$$

must be equal to 1. The remaining eigenvalues of P are exactly the other diagonal entries of U :

$$\lambda_i(P) = U_{ii}, \quad i = 1, \dots, n-1,$$

up to a relabeling of indices. In particular, the full multiset of eigenvalues of P can be described as

$$\{1\} \cup \{U_{11}, \dots, U_{nn}\} \setminus \{\text{one diagonal entry that is shifted to } 1\}.$$

Equivalently, we may write: the eigenvalues of P consist of 1 together with the diagonal entries of U , except for one diagonal entry that has been shifted to 1 by the rank-one perturbation.

5. Calculate the spectral gap of this independent Metropolis–Hastings chain.

Solution: From part (4), the eigenvalues of P consist of

$$1 \quad \text{and} \quad U_{11}, \dots, U_{nn},$$

with exactly one of the diagonal entries U_{ii} shifted to 1 by the rank-one perturbation. In particular, all *nontrivial* eigenvalues (those different from 1) are among the diagonal entries of U :

$$\{\lambda : \lambda \neq 1\} \subset \{U_{11}, \dots, U_{nn}\}.$$

Recall from part (2) that

$$U_{ii} = \sum_{j>i} p_j \left(1 - \frac{w_j}{w_i}\right), \quad i = 1, \dots, n,$$

where $w_1 \geq \dots \geq w_n$ by assumption. For $j > i$ we have $w_j \leq w_i$, so $1 - w_j/w_i \geq 0$ and hence each $U_{ii} \geq 0$.

For a reversible Markov chain with transition matrix P , the spectral gap is defined as

$$\gamma = 1 - \max\{|\lambda| : \lambda \text{ is an eigenvalue of } P, \lambda \neq 1\}.$$

Using the description above and the fact that all nontrivial eigenvalues are non-negative, this becomes

$$\gamma = 1 - \max\{\lambda : \lambda \neq 1\} = 1 - \max_{i: U_{ii} \neq 1} U_{ii}.$$

Equivalently, we can write

$$\gamma = 1 - \max_i U_{ii} = 1 - \max_{1 \leq i \leq n} \sum_{j>i} p_j \left(1 - \frac{w_j}{w_i}\right),$$

where the maximum is taken over all indices i such that U_{ii} is a nontrivial eigenvalue (i.e. not shifted to 1). In terms of the proposal probabilities (p_j) and weights (w_j), this expression gives the spectral gap of the independent Metropolis–Hastings chain.